

SAMPLE DATA DOCUMENTATION

Project Name

Postman Collection to Pytest Migrator

1. Purpose of Sample Data

The purpose of this sample data is to demonstrate how the application converts a Postman Collection into an executable Pytest automation script.

The sample collection contains API requests and Postman assertions. The system parses the collection, validates the structure, extracts assertions, and generates corresponding Pytest code.

This sample data is used for functional testing, validation, and demonstration purposes.

2. Sample Input Collection

Collection Information

Collection Name: Pytest Migrator Verification Collection

Schema Version: Postman Collection v2.1

API Type: REST API

Endpoint: <https://jsonplaceholder.typicode.com/users/1>

HTTP Method: GET

Input Postman Collection JSON

```
{
  "info": {
    "name": "Pytest Migrator Verification",
    "schema": "https://schema.getpostman.com/json/collection/v2.1.0/collection.json"
  },

```

```
"item": [  
  {  
    "name": "Get User",  
    "request": {  
      "method": "GET",  
      "url": {  
        "raw": "https://jsonplaceholder.typicode.com/users/1"  
      }  
    },  
    "event": [  
      {  
        "listen": "test",  
        "script": {  
          "exec": [  
            "pm.response.to.have.status(200);",  
            "pm.expect(pm.response.json().id).to.eql(1);",  
            "pm.expect(pm.response.json().username).to.eql('Bret');"  
          ]  
        }  
      }  
    ]  
  }  
]
```

3. Data Extraction Process

The parser performs the following operations:

Step 1: Collection Validation

The uploaded JSON file is validated against Postman Collection v2.1 standards.

Validated Components:

- Collection metadata
- Request definitions
- Event definitions
- Assertion blocks

Step 2: API Extraction

Extracted API Details:

Field	Value
API Name	Get User
Method	GET
URL	https://jsonplaceholder.typicode.com/users/1
Headers	None
Query Parameters	None

Step 3: Assertion Extraction

The assertion extraction engine identifies:

Postman Assertion	Extracted Meaning
pm.response.to.have.status(200)	Verify status code equals 200
id equals 1	Verify user ID
username equals Bret	Verify username

Total Assertions Extracted: 3

4. Expected Output

After processing, the system generates a Pytest script.

Generated Pytest Script

```
import pytest
import requests

def test_get_user():

    response = requests.get(
        "https://jsonplaceholder.typicode.com/users/1"
    )

    assert response.status_code == 200

    assert response.json()["id"] == 1
```

```
assert response.json()["username"] == "Bret"
```

5. Generated Recommendations

The recommendation engine automatically generates:

Positive Test Recommendation

Verify that the API returns HTTP 200 for valid requests.

Negative Test Recommendation

Verify behavior when an invalid user ID is supplied.

Boundary Test Recommendation

Verify behavior for maximum supported user identifiers.

Security Recommendation

Validate response schema and sensitive data exposure.

6. Generated Test Cases

Test Case	Type	Expected Result
Validate User Retrieval	Positive	Status Code 200
Invalid User ID	Negative	Error Response
Boundary User Validation	Boundary	Stable Response
Response Schema Validation	Security	Schema Compliance

7. Conversion Summary

Metric	Value
Collections Uploaded	1

Metric	Value
APIs Extracted	1
Assertions Extracted	3
Pytest Scripts Generated	1
Recommendations Generated	4
Test Cases Generated	4

8. Verification Result

The generated Pytest script was manually reviewed and verified.

Verification Checks:

- ✓ HTTP Request Generated
- ✓ Status Assertion Converted
- ✓ JSON Assertions Converted
- ✓ Python Syntax Valid
- ✓ Pytest Compatible
- ✓ Downloadable Script Generated

9. Conclusion

The sample data demonstrates the complete workflow of the Postman Collection to Pytest Migrator.

The application successfully:

- Parses Postman collections
- Extracts API details
- Extracts assertions
- Generates executable Pytest code
- Produces recommendations
- Generates test cases

This validates the correctness and effectiveness of the proposed solution.